Fiscal Year:	FY 2007	**Task Last Updated:**	FY 07/30/2007
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Project Title:	Mechanosensing and Mechanotransduction by the Na-H exchanger NHE1		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	05/01/2004	**End Date:**	04/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NCE to 4/30/2007, per A. Chu 6/23/06		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CC53A		
Performance Goal No.:			
Performance Goal Text:			
	In single-cell organisms plasma membrane Na-H exchangers (NHEs) function as osmoregulators. In multicellular organisms, we propose that the functions of NHEs have evolved to include mechanosensing and mechanotransduction as an adaptive response to the specialized needs of mechanically integrated tissues. One component of this adaptive response is that in multicellular organisms NHEs are structurally linked to the actin-based cytoskeleton. In metazoans, NHEs catalyze an electroneutral exchange of extracellular Na+ and intracellular H+, which regulates intracellular pH and volume homeostasis, and they act as plasma membrane anchors of the actin cytoskeleton, which regulates cells shape and adhesion. Conserved responses of metazoan cells to gravity-generated forces include changes in cell shape, cytoskeleton organization, and cell adhesion. The objective of this proposal is determine the role of the uniquitously expressed Na-H exchanger NHE1 in gravity-generated responses by investigating the hypothesis that NHE1 is a critical		

Task Description:	determinant in the sensing and transducing of mechanical strain. Because mechanical forces are important regulators of connective tissue homeostasis, and fibroblasts sense force-induced deformations (strains) in the extracellular matrix we will use fibroblast models developed in our lab that express mutant NHE1 with impaired ion translocation or cytoskeletal anchoring. In Aim 1, the role of NHE1 in mechanosensing will be investigated by studying NHE1 activity in whole cells and in excised membrane patches in response to mechanical loading, by determining whether cytoskeletal anchoring by NHE1, cell adhesion, and integrin activation are necessary for NHE1 activation by mechanical loading. In Aim 2, the function of NHE1 in mechanotransduction will be investigated by using conventional microscopy and atomic force microscopy to determine whether NHE1 activity or cytoskeletal anchoring are necessary for dynamic changes in cell topology, and in remodeling of the actin cytoskeleton and cell-substrate adhesions in response to mechanical loading. Processes controlling cytoskeleton remodeling, including de-novo actin polymerization and the activation of Rho family GTPases will be determined in response to mechanical loading, and the dependence of NHE1 in this response will be evaluated. These studies explore a previously unrecognized role of NHE1 in mechanosening and mechanotransduction, and they are designed to further our understanding of how gravity-generated forces regulate cell functions. The overall objective is to identify molecular components of a mechanotransduction unit in fibroblasts. In contrast to the wealth of information known about how cells sense chemical stimuli, little is known about how cells sense mechanical forces and transduce this information into biochemical events and cell responses. An understanding of this process will likely provide signficant insight into how mechanical forces modulate cell and tissue function, morphogenesis, differentiation, and remodeling.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We identified the ubiquitously expressed Na-H exchanger NHE1 as a necessary component of a mechanotransduction unit in fibroblasts. This finding is significant in understanding how cells sense and respond to mechanical force, which is a important for determining how mechanical force drives normal processes such as development, tissue organization, wound-healing, and cell proliferation and migration, and contributes to pathological conditions such as atherosclerosis and fibrosis.
Task Progress:	The study of mechanotransduction is an exciting new direction for our research program. We previously developed most of the cell models described in our research plan, and have considerable expertise in studying cytoskeletal dynamics and activities of Rho family GTPases in response to growth factors and hormones. The first year of our NASA award was spent developing a uniaxial loading device with NASA Ames researchers Dr. Eduardo Almeida and Nancy Searby, and optimizing conditions for using these devices to determine responses to cell stretch by using biochemical assays and live-cell imaging. New data obtained since our previous progress report include the following: Aim 1. Our previous reports included data showing that mechanical force increases intracellular pH and that this is blocked by pharmacological agents inhibiting activity of the Na-H exchanger, NHE1. We also showed that mechanical force increases NHE1 activity and phosphorylation. One objective we accomplished during the current funding period was to confirm that activation of NHE1 by mechanical load occurs independently of activation of integrin cell adhesion receptors. A current view is that because integrins bind to extracellular matrix, which is deformed during mechanical stretch, their activation is necessary for mechanosensitivity. However, we found that the magnitude of NHE1 activity stimulated by mechanical load is similar when cells are plated on a synthetic poly-L-lysine substrate that does not activate integrins compared with cells plated on the matrix protein fibronectin, which activates integrins. We also found that NHE1 is activated by mechanical force in F9 cells engineered to lack beta1-integrin, which is necessary for binding to fibronectin matrices. These data indicate that activation of integrin receptors is not necessary for increased NHE1 activity and intracellular pH. Aim 2. We completed the objectives of Aim 2 to determine whether NHE1 is necessary for remodeling of the actin cytoskeleton and focal adhesions in response to mechanical load. Reorganization of actin filaments and an increase in actin polymerization are conserved responses to mechanical force. We found that cells having NHE1 activity but not mutant cells lacking NHE1, mechanical load induces an increase in the assembly, bundling and organization of actin filaments. We completed biochemical experiments to understand how NHE1 promotes actin assembly in response to mechanical load. We found that Arp2/3, a protein complex that functions as a key stimulator of de-novo actin assembly is phosphorylated in response to mechanical load in cells having NHE1 activity but not mutant cells lacking NHE1. We also confirmed that phosphorylation of Arp2/3 is necessary for its stimulation of actin assembly, which suggests that an NHE1-dependent phosphorylation of Arp2/3 is necessary for increased actin assembly in response to mechanical load. Our studies on focal adhesions were completed in previous funding periods and confirmed that focal adhesion assembly and size, and activation of the focal adhesion kinase FAK increase in response to mechanical load in cells having NHE1 activity but not mutant cells lacking NHE1. These data suggest that NHE1 functions in mechanotransduction because it is necessary for conserved responses to mechanical load, including actin assembly, activation of FAK, and the assembly of focal adhesions. We currently are preparing a manuscript describing these data in the context of being a first report showing that NHE1 plays a central role in mechanotransduction. Additionally, these findings were used as preliminary data in a grant proposal on mechanotransduction that was recently funded by National Institutes of Health (grant number GM71526). Hence funding from our NASA award supported significant new findings from our laboratory and facilitated the expansion of these studies through NIH funding.
Bibliography Type:	Description: (Last Updated: 07/31/2007)
Abstracts for Journals and Proceedings	Mackley JR, Frantz C, LeClaire L, Almeida E, Barber, DL. "Mechanosensing and mechanotransduction by the Na-H exchanger NHE1 for actin filament assembly." Annual Meeting of the American Society for Cell Biology, December 2006. Mol Biol Cell 2006;17(Suppl):1564. , Dec-2006